Data Science with Python
Ticket Out the Door
Set 23 Ticket Out The Door

Name ____ Period ______

Skill 23.1 Exercise 1

The DataFrame *df* below contains information on products sold at a hardware store.

	Product ID	Description	Cost to Manufacture	Price
0	1	3 inch screw	0.5	0.75
1	2	2 inch nail	0.1	0.25
2	3	hammer	3.0	5.5
3	4	screwdriver	2.5	3.0

Add a column to *df* called *Sold in Bulk?*, which indicates if the product is sold in bulk or individually. The final table should look like the one below,

Product ID	Product Description	Cost to Manufacture	Price	Sold in Bulk?
1	3 inch screw	0.50	0.75	Yes
2	2 inch nail	0.10	0.25	Yes
3	hammer	3.00	5.50	No
4	screwdriver	2.50	3.00	No

Add a column to *df* called *Is taxed?*, which indicates whether or not to collect sales tax on the product. It should be 'Yes' for all rows

Add a column to *df* called *Margin*, which is equal to the difference between the *Price* and the *Cost to Manufacture*.

Data Science with Python
Ticket Out the Door
Set 23 Ticket Out The Door

Name ____ Period ______

Skill 23.2 Exercise 1

Refer to the DataFrame below.

	Name	Email
0	JOHN SMITH	john.smith@gmail.com
1	JANE DOE	jdoe@yahoo.com
2	JOE SCHMO	joeschmo@hotmail.com

Apply the function *lower* to all names in column *Name* in *df*. Assign these new names to a new column of *df* called *Lowercase Name*. The final DataFrame should look like the one below,

	Name	Email	Lowercase Name
0	JOHN SMITH	john.smith@gmail.com	john smith
1	JANE DOE	jdoe@yahoo.com	jane doe
2	JOE SCHMO	joeschmo@hotmail.com	joe schmo

Skill 23.3 Exercise 1

Suppose we want to pay workers time-and-a-half for overtime (any work above 40 hours per week). The following function will convert the number of hours into time-and-a-half hours using an if statement:

```
def myfunction(x):
    if x > 40:
        return 40 + (x - 40) * 1.50
    else:
        return x
```

Rewrite the function above as a lambda function named *adj_hours*

Name ____ Period _____

Skill 23.4 Exercise 1

Below is a DataFrame *df* that contains information on employees, including their hourly wage and their hours worked.

	Unnamed: 0	id	name	hourly_wage	hours_worked
0	0	10310	Lauren Durham	19	43
1	1	18656	Grace Sellers	17	40
2	2	61254	Shirley Rasmussen	16	30
3	3	16886	Brian Rojas	18	47
4	4	89010	Samantha Mosley	11	38
5	5	87246	Louis Guzman	14	39
6	6	20578	Denise Mcclure	15	40
7	7	12869	James Raymond	15	32

Apply the lambda function you wrote a above to the *hours_worked* column to calculate the time-and-a-half hours. Store the result in a new column called *total_hours*.

Calculate the total earned for each employee by multiplying the *hourly_wage* column by the *hours_worked* column. Store the result in a new column called *earnings*

Skill 23.5 Exercise 1

The DataFrame *df* contains data about movies from IMDb. Right now, the column names are in lower case, and are not very descriptive. Modify *df* using the *columns* attribute to make the following changes to the columns,

Old	**New**
id	ID
name	Title
genre	Category
year	Year Released
rating	IMDB_Rating

Data Science with Python
Ticket Out the Door
Set 23 Ticket Out The Door

Name ____ Period _____

Skill 23.5 Exercise 2

If we didn't know that *df* was a table of movie IMDB ratings, the columns *name* and *rating* might be confusing. To clarify, rename *name* to *movie_title* and *rating* to *IMDB_rating*   Use the keyword inplace=True so that you modify *df* rather than creating a new DataFrame!

	id	name	genre	year	rating
0	1	Avatar	action	2009	7.9
1	2	Jurassic World	action	2015	7.3
2	3	The Avengers	action	2012	8.1
3	4	The Dark Knight	action	2008	9.0
4	5	Star Wars: Episode I - The Phantom Menace	action	1999	6.6